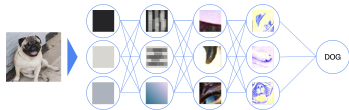


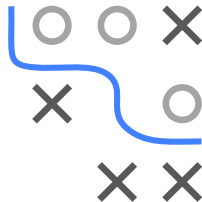
Introduction to Machine Learning

Neural Networks In a Nutshell

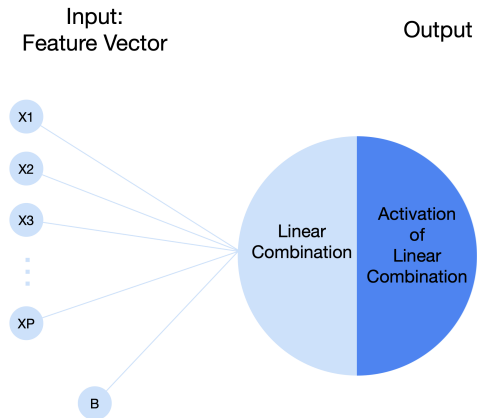


Learning goals

- Know basic computational unit
- Know basic architecture
- Understand Learning in NNs

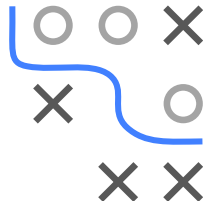


BASIC COMPUTATIONAL UNIT: PERCEPTRON I



Output differs depending on activation function:

- **Identity function:**
Perceptron represents linear regression
- **Logistic function:**
Perceptron represents logistic regression
- Other activation functions possible



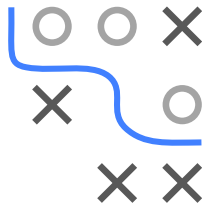
BASIC ARCHITECTURE OF NN I

Hidden Layers:

- Output of hidden units serves as input for units in next layer
- Too many hidden layers or too many units per layer can lead to overfitting

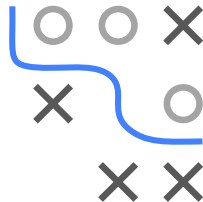
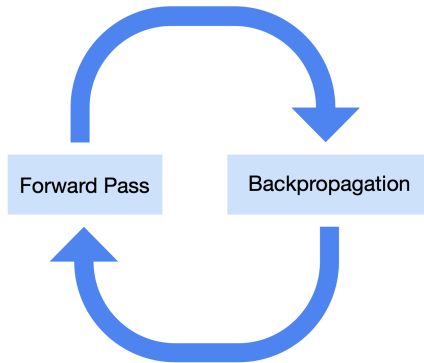
Output Layer:

- Number of output units depend on task
- Different activation functions for output layer and hidden layers possible

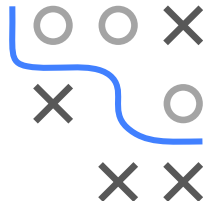
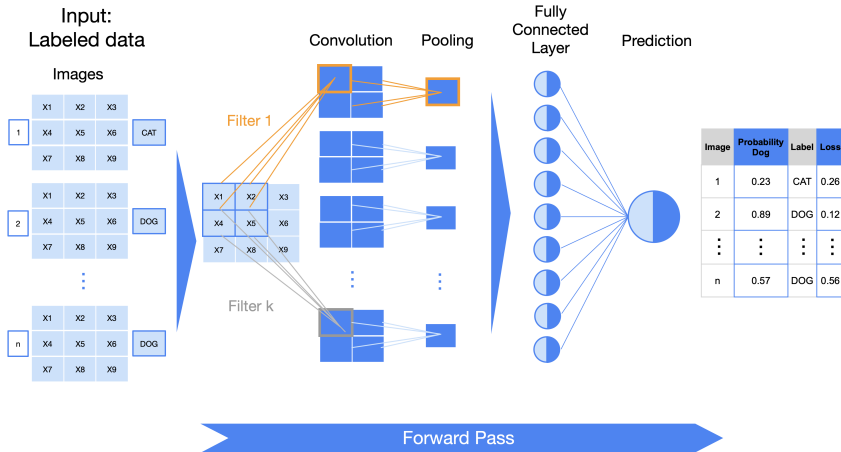


LEARNING - IMAGE CLASSIFICATION TASK I

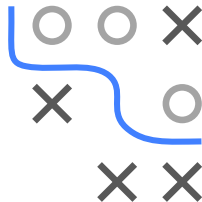
For each Training Iteration:



LEARNING - IMAGE CLASSIFICATION TASK I



Compute update of each weight by backpropagation



PREDICTION - IMAGE CLASSIFICATION TASK I

